

Task Description (type, audience, purpose)

For this task, I am working on developing a prompt to process student feedback from one of our course sessions. My goal is to generate a quick, at-a-glance summary (1 page or less) of key metrics, themes, and areas for improvement. However, I decided to adopt an intermediate goal based on some initial work that one of my colleagues had done: a prompt which will ask the LLM to categorize **all** student responses into themes (as one would in qualitative research) to make sure that the results are more easily verifiable. Since all the quotes are there, I can at least spot check whether they exist in the original version and whether the LLM has generated a reasonable set of themes.

Developing a system to do this for all course sessions would be useful because this course has ~35 sessions, and this could then serve as a quick temperature check for what changes we might need to make for the session the next time it runs. Eventually, developing a separate prompt/workflow for analyzing themes from all the session feedback together would be useful, but I'm starting small.

The target audience for this would be our course team, made up of the course directors and instructional designers. Though the data is somewhat sensitive, the good news is that all of these people already have access to feedback data, so data sensitivity is only a concern for what goes into the LLM and not for the end users.

Tools Used

I decided to use Claude Opus 4.6 (the highest-resource version of Claude available to us at U of A through the Open WebUI portal) for my initial phase, which involved prompt drafting and refinement before actually attempting the task with my test data.

I'm wondering whether it would make sense to continue using Opus once I actually do the analysis, but I suppose it might make sense to compare Haiku, Sonnet, and Opus to see how they stack up. Since this would be a repeated task, it might be nice to use a lower-resource version, but that's only if it gives reasonably good results.

Data Sensitivity

For this task, all student identifiers (name, e-mail address, etc.) will be deleted from the spreadsheet before submission. I will also use a find + replace operation in Excel to use pseudonyms for all course instructors as appropriate. I'll then have to de-pseudonym-ize

(there's a fun word) the data once we have the final output, that that's also easily accomplished with a find + replace operation.

Normal Time for This Task

I've never done this task by hand myself because there's not enough time to do it for all of our sessions. I would anticipate that doing a thematic analysis for one session would probably take at least 3-4 hours.

Actual Time This Session

Prompt Sequence and Associated Output Summary

Note: due to length, I will upload prompts in a separate file.

Part 1: Prompt Development Sequence

Prompt 1 (general instructions)

I will be sharing a long prompt intended to work with a large language model to generate a report on student feedback from a university course session. In the subsequent prompt, disregard any instructions. Instead, I want you to act as an expert LLM prompt engineer to analyze the prompt for any areas of improvement. Ask any important clarifying questions and offer feedback on areas where you think the prompt could be improved, whether that is adding, removing, or editing.

Output Summary 1

[output was a general go-ahead response, basically "sure, and here are the categories I'll consider"]

Prompt 2 (the prompt on which I was seeking feedback)

The data contained in this spreadsheet is a set of student feedback from a course session in a college of veterinary medicine. Each row represents a student comment. Each column is a distinct question and should be analyzed separately.

The first sheet is feedback on the session prework. Here are the questions asked and the columns they correspond to on the spreadsheet.

- Q1 Hours: How many hours did the prework in this session take you to complete?

- This is a closed response item. Respondents can only choose one answer.
The three answer choices are:
 - 0 to 2 hours
 - 2 to 3 hours
 - more than 3 hours
- Q2 Understand: What materials or content in the prework (if any) do you understand the most?
 - This is an open response (free text) item.
- Q3 Struggle: What materials or content in the prework (if any) did you struggle to understand?
 - This is an open response (free text) item.
- Q4 LOs Covered: Did this prework cover the learning objectives listed for this session?
 - This is a closed response item. Respondents can only choose one answer.
The four answer options are:
 - Definitely not
 - Probably not
 - Probably yes
 - Definitely yes
- Q5 Questions: Do you have any other questions or feedback?
 - This is an open response (free text) item.

The second sheet is feedback on how the session went. Here are the questions asked and the columns they correspond to on the spreadsheet.

- Q1 Content & LO: Did the content in the session adequately support the learning objectives listed for this session?
 - This is a closed-response item. Respondents can only choose one answer.
The answer choices are:
 - Definitely not
 - Probably not
 - Probably yes
 - Definitely yes
- Q2 Muddy: What is still a muddy point?
 - This is an open response (free text) item.
- Q3 Takeaway: What is your biggest takeaway from the session?
 - This is an open response (free text) item.
- Q4 What Worked: What worked particularly well in the session?

- This is a closed response item. It is a multi-select question, meaning respondents can choose more than one response. The answer choices are:
 - Instructor facilitation
 - Hands on activities
 - Team discussion
 - Class discussion
 - Pework
- Q5 Suggestions: Do you have any suggestions for how we could improve the session in the future?
 - This is an open response (free text) item.

Act as an expert evidence-based qualitative researcher, instructional design expert, and faculty developer. I would like you to generate a report based solely on the data in this spreadsheet.

For the prework data (first sheet of the spreadsheet), please use the following format:

- Q1 Hours: Create a table with the number of responses for each item. Include both the number of students and the percentage of students that gave that response.
- Q2 Understand, Q3 Struggle, and Q5 Questions: Create a set of general qualitative themes from the data provided for each question. Also create subthemes if appropriate. For each theme and subtheme identified, show all quoted comment(s) that were categorized into that theme. Also identify the number of comments that constituted that theme/subtheme. Each row should be represented even if it does not create a subtheme; it can be put under the theme of Other Comments, but group like comments together first. List the Other Comments category after the other themes and subthemes.
- Q4 LOs Covered: Create a table with the number of responses for each item. Include both the number of students and the percentage of students that gave that response.

For the post-session data (second sheet of the spreadsheet), please use the following format:

- Q1 Content & LO: Create a table with the number of responses for each item. Include both the number of students and the percentage of students that gave that response.
- Q2 Muddy: Create a set of general qualitative themes from the data provided for each question. Also create subthemes if appropriate. For each theme and subtheme identified, show all quoted comment(s) that were categorized into that

theme. Also identify the number of comments that constituted that theme/subtheme. Each row should be represented even if it does not create a subtheme; it can be put under the theme of Other Comments, but group like comments together first. List the Other Comments category after the other themes and subthemes.

- Q3 Takeaway: Create a set of general qualitative themes from the data provided for each question. Also create subthemes if appropriate. For each theme and subtheme identified, show all quoted comment(s) that were categorized into that theme. Also identify the number of comments that constituted that theme/subtheme. Each row should be represented even if it does not create a subtheme; it can be put under the theme of Other Comments, but group like comments together first. List the Other Comments category after the other themes and subthemes. However, some students may put suggestions or observations in this question that might belong better in Q5 Suggestions. In this question, please only include items which relate to the content of the session. For any responses that are suggestions or observations, please include those in your analysis of the data from Q5 as instructed below.
- Q4 What Worked: Create a table with the total number of times that each responses was selected. Include both the number of students that gave that response. The total number of responses may exceed the number of respondents, but since this is a multi-select question, that's not a problem.
- Q5 Suggestions: Create a set of general qualitative themes from the data provided for each question. Also create subthemes if appropriate. For each theme and subtheme identified, show all quoted comment(s) that were categorized into that theme. Also identify the number of comments that constituted that theme/subtheme. Each row should be represented even if it does not create a subtheme; it can be put under the theme of Other Comments, but group like comments together first. List the Other Comments category after the other themes and subthemes.

Do not include any blank responses or responses like "N/A," "none," or "no" in your analysis. Ignore any blank rows. Remove HTML tags entirely before processing; do not treat them as content boundaries.

Output Summary 2

- set of 5 bulleted strengths
- 10 areas for improvement:
 - handling of non-substantive responses

- HTML tag instructions
- appropriately specifying thematic analysis methodology
- clarifying quoting conventions
- appropriate instructions for handling multi-select question data
- providing instruction on structure & headers for report
- providing information on tone and audience
- optional: asking for cross-question analysis
- clarifying what “every row should be represented” means
- clarifying how to handle ambiguous closed-response data
- 6 clarifying questions & suggestions
 - method of spreadsheet upload
 - how to format qualitative themes
 - how to handle duplicate/near-duplicate comments
 - desired output form
 - suggestion to maybe add session titles and/or dates
 - how to organize ‘Other Comments’ category

Prompt 3

Clarifying Question 1: Why would I need to instruct the LLM to parse the spreadsheet file? What does this mean, and how would an LLM likely interpret this instruction?

Output Summary 3

[The model changed its mind and said in retrospect, my prompt is clear enough that that instruction is not necessary.]

Prompt 4

No thank you. I’m going to upload the updated prompt, and if you have any additional input, I’d be happy to consider it.

Output Summary 4

[generic go-ahead]

Prompt 5

The data contained in this spreadsheet is a set of student feedback from a course session in a college of veterinary medicine. Each row represents a student comment. Each column is a distinct question and should be analyzed separately.

The first sheet is feedback on the session prework. Here are the questions asked and the columns they correspond to on the spreadsheet.

- Q1 Hours: How many hours did the prework in this session take you to complete?
 - This is a closed response item. Respondents can only choose one answer.

The three answer choices are:

 - 0 to 2 hours
 - 2 to 3 hours
 - more than 3 hours
- Q2 Understand: What materials or content in the prework (if any) do you understand the most?
 - This is an open response (free text) item.
- Q3 Struggle: What materials or content in the prework (if any) did you struggle to understand?
 - This is an open response (free text) item.
- Q4 LOs Covered: Did this prework cover the learning objectives listed for this session?
 - This is a closed response item. Respondents can only choose one answer.

The four answer options are:

 - Definitely not
 - Probably not
 - Probably yes
 - Definitely yes
- Q5 Questions: Do you have any other questions or feedback?
 - This is an open response (free text) item.

The second sheet is feedback on how the session went. Here are the questions asked and the columns they correspond to on the spreadsheet.

- Q1 Content & LO: Did the content in the session adequately support the learning objectives listed for this session?
 - This is a closed-response item. Respondents can only choose one answer.

The answer choices are:

 - Definitely not
 - Probably not
 - Probably yes
 - Definitely yes
- Q2 Muddy: What is still a muddy point?
 - This is an open response (free text) item.
- Q3 Takeaway: What is your biggest takeaway from the session?
 - This is an open response (free text) item.

- Q4 What Worked: What worked particularly well in the session?
 - This is a closed response item. It is a multi-select question, meaning respondents can choose more than one response. The answer choices are:
 - Instructor facilitation
 - Hands on activities
 - Team discussion
 - Class discussion
 - Prework
- Q5 Suggestions: Do you have any suggestions for how we could improve the session in the future?
 - This is an open response (free text) item.

Act as an expert evidence-based qualitative researcher, instructional design expert, and faculty developer. I would like you to generate a report based solely on the data in this spreadsheet. The report should contain a header with the session number. The session number will be in the title of the uploaded spreadsheet. Structure the report with clear section headers: 'Prework Feedback' and 'Post-Session Feedback,' with sub-headers for each question. Begin with a brief executive summary (3-5 sentences) highlighting key findings across both sheets.

The audience for this report is the course team (course directors and instructional designers). Do not include interpretive commentary or recommendations beyond the data.

Qualitative Analysis Process: Create a set of general qualitative themes from the data provided for each question. Also create subthemes if appropriate. At least 2 comments are needed to create a theme. Single comments that do not fit a theme should be placed in an 'Other Comments' section. List the Other Comments category after the other themes and subthemes. Theme names should be concise descriptive phrases (2-7) words. For each theme and subtheme identified, show all quoted comment(s) that were categorized into that theme, including duplicates and near-duplicates. Also identify the number of comments that constituted that theme/subtheme. Every substantive student response must be categorized into exactly one theme or subtheme, or placed in 'Other Comments.' No substantive response should be omitted from the analysis. If a single student response addresses multiple themes, the relevant portion may be quoted under each applicable theme. Indicate when a quote is an excerpt by using [...] for omitted portions. Format all qualitative analyses as nested lists.

Quantitative Analysis Process: Create a table with the number of responses for each item. Include both the number of students and the percentage of students that gave that

response. Calculate percentages based on the number of valid responses (excluding blanks). If blanks exist, note the number of non-responses below the table.

For the prework data (first sheet of the spreadsheet), please use the following format:

- Q1 Hours: Use the Quantitative Analysis Process included earlier in this prompt.
- Q2 Understand, Q3 Struggle, and Q5 Questions: Use the Qualitative Analysis Process included earlier in this prompt.
- Q4 LOs Covered: Use the Quantitative Analysis Process included earlier in this prompt.

For the post-session data (second sheet of the spreadsheet), please use the following format:

- Q1 Content & LO: Use the Quantitative Analysis Process included earlier in this prompt.
- Q2 Muddy: Use the Qualitative Analysis Process included earlier in this prompt.
- Q3 Takeaway: Use the Qualitative Analysis Process included earlier in this prompt. However, some students may put suggestions or observations in this question that might belong better in Q5 Suggestions. In this question, please only include items which relate to the content of the session. For any responses that are suggestions or observations, please include those in your analysis of the data from Q5 as instructed below.
- Q4 What Worked: Use the Quantitative Analysis Process included earlier in this prompt. The total number of responses may exceed the number of respondents, but since this is a multi-select question, that's not a problem.
- Q5 Suggestions: Use the Qualitative Analysis Process included earlier in this prompt.

Exclude from your analysis any non-substantive responses that do not provide specific content, including but not limited to: 'N/A,' 'none,' 'no,' 'nothing,' 'nope,' 'all good,' 'I'm good,' 'not really,' or similar. Ignore any blank rows. Strip all HTML tags but preserve the text content within them. Convert HTML entities (e.g., & → '&') to their plain-text equivalents.

What Changed in Prompt 5

- updated language regarding non-substantive responses
- clarified instructions on how to handle HTML tags
- clarified that themes require 2+ comments and should be descriptive phrases
- specified what to do with responses that touch on multiple themes

- clarified how to calculate percentages for the multi-select question
- provided instructions on headers and subheaders to use in the report
- included information on intended audience
- clarified what “each row should be represented” means
- provided instructions on how to format qualitative themes
- provided instructions on how to handle duplicate/near-duplicate comments
- provided instructions on how to organize the “Other Comments” category

Output Summary 5

The LLM said that the prompt was much improved.

5 minor issues:

- still need to clarify how to handle multi-select question responses
- improving instructions on how to title the report if the LLM can’t read the titles of uploaded files (evidently some can, and some might not be able to?)
- what to do if there are no themes
- providing an example of how to format nested lists
- minor formatting fix in one of the text lines

It also offered to create a **condensed version of this prompt for token efficiency**.

Prompt 6

Yes, I would like you to generate a condensed version for token efficiency.

[It noted after providing the output below that this decreased the word count from about 950 to about 480, a reduction of 48%!]

Output 6 (non-summary)

You are an expert evidence-based qualitative researcher, instructional design expert, and faculty developer. Generate a report based solely on the attached spreadsheet data, which contains student feedback from a course session in a college of veterinary medicine. The audience is the course team (course directors and instructional designers). Do not include interpretive commentary or recommendations beyond the data.

Report Structure:

- Header with session number (found in the spreadsheet filename; use '[Session Number]' if inaccessible)
- Executive summary (3-5 sentences) highlighting key findings across both sheets

- Section headers: 'Prewrite Feedback' and 'Post-Session Feedback,' with sub-headers for each question

Data Handling Rules:

- Each row = one student; each column = one question (analyze separately)
- Exclude non-substantive responses (e.g., 'N/A,' 'none,' 'no,' 'nothing,' 'nope,' 'all good,' 'I'm good,' 'not really,' or similar) and blank rows
- Strip HTML tags (preserve text within); convert HTML entities to plain text (e.g., & → '&')

Quantitative Process (closed-response items):

Create a table showing each answer choice, number of responses, and percentage of valid responses (excluding blanks). Note non-response count below the table if blanks exist.

Qualitative Process (open-response items):

Create concise themes (2-7 word names) requiring at least 2 comments. Single comments go in 'Other Comments' (listed last). If no themes meet the threshold, state this and list all under 'Other Comments.' Create subthemes where appropriate. For each theme/subtheme, list all quoted comments (including duplicates/near-duplicates) and the comment count. Every substantive response must appear in exactly one theme/subtheme or 'Other Comments.' If a response spans multiple themes, excerpt relevant portions under each (use [...] for omissions). Format as nested lists:

- **Theme Name** (n = X)
 - *Subtheme Name* (n = X)
 - "Student quote"

Sheet 1 — Prewrite Feedback:

- Q1 Hours (closed; choices: 0-2 hours / 2-3 hours / more than 3 hours): Quantitative Process
- Q2 Understand (open): Qualitative Process
- Q3 Struggle (open): Qualitative Process
- Q4 LOs Covered (closed; choices: Definitely not / Probably not / Probably yes / Definitely yes): Quantitative Process
- Q5 Questions (open): Qualitative Process

Sheet 2 — Post-Session Feedback:

- Q1 Content & LO (closed; choices: Definitely not / Probably not / Probably yes / Definitely yes): Quantitative Process
- Q2 Muddy (open): Qualitative Process
- Q3 Takeaway (open): Qualitative Process — include only content-related responses. Move any suggestions/observations to Q5 analysis.
- Q4 What Worked (multi-select; choices: Instructor facilitation / Hands on activities / Team discussion / Class discussion / Prework): Quantitative Process — report percentage of respondents selecting each option (percentages may exceed 100%).
- Q5 Suggestions (open): Qualitative Process — include any suggestions re-routed from Q3.

Part 2: Using the Prompt

Prompt 1

I used the prompt included in Output 6 above, uploading a pseudonymized version of the session feedback from one of our courses.

Output Summary 1

As expected, the model generated a very long summary, since it needed to integrate all student comments. When pasted into Microsoft Word, the report was 23 pages long.

This is much longer than I would like the eventual products of this process to be, but I think that this is a good start because it provides an output that should be straightforward to verify for intrinsic hallucinations (against the original prompt spreadsheet). Given how long I've already spent on this task, I'm going to call this good progress for now and plan to return to it later in the course or once we're done.

Disclosure Decision

Assuming we were to adopt this process, I think it would be appropriate to let students know that we're planning to use AI tools to analyze feedback and why we're doing it. Writing a thoughtful rationale would be helpful because I'm sure some of the students would be somewhat skeptical ("Why can't you just read it yourself?"). The [U of A AI Tools & Data Use Guide](#) suggests disclosing using the use of AI in creating content, so this is probably a good idea.